Viability-Preserving Passive Torque Control for Mobile Base Manipulator

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***Abstract*—This report presents an extension of the Viability-Preserving Passive Torque Control (VPP-TC) framework to a mobile manipulator system consisting of a 7-DOF robotic arm mounted on a 2-DOF omnidirectional mobile base. The proposed framework ensures safe, collision-free trajectory tracking by strictly enforcing constraints on joint limits, self-collisions, and external obstacles through the utilization of precomputed viability sets and instantaneous Quadratic Programming (QP). Unlike reactive safety filters, VPP-TC guarantees infinite-horizon safety by maintaining the system state within learned viable sets, parameterized using Transformer architectures for self-collision and Bernstein polynomials for external obstacles. We perform a comparative analysis against a state-of-the-art Control Barrier Function (CBF) based baseline, specifically the Constrained Passive Interaction Control (CPIC) method. Experimental results demonstrate that the VPP-TC framework significantly outperforms the CBF baseline in computational efficiency, achieving a control loop rate of 110 Hz compared to 47 Hz. Furthermore, VPP-TC exhibits superior numerical stability and smoother control outputs, effectively eliminating the jitter observed in the Hessian-dependent CBF method.**

I. Introduction

Safety-critical control for manipulator and mobile-manipulator systems has been approached from various directions [2][3][4][5], including optimization-based Control Barrier Function (CBF) methods, whole-body QP control, and more recently viability- and learning-based safe sets. Constrained Passive Interaction Control (CPIC) [1] is a framework that explicitly combines passivity and safety for torque-controlled manipulators. CPIC starts from a dynamical systems-based impedance controller and enforces joint limits, self/external collision via a relaxed hierarchical CBF-QP. Safety constraints are formulated in joint space from $\mathcal{C}^2$ data-driven self- and external collision boundary functions, learned by sampling the robot's joint space [1]. At each control step, a QP modifies the nominal passive torque command to satisfy multiple, possibly conflicting CBF constraints.

Classical passivity-based torque controllers, such as impedance controllers, provide robustness and stability under interaction but do not by themselves guarantee satisfaction of kinematic or collision constraints. Zhang et al. address this limitation with Viability-Preserving Passive Torque Control on a fixed base manipulator (VPP-TC) [6]. VPP-TC employs viability theory to pre-compute safe sets in the joint position–velocity space. These viable sets, constructed via data-driven and analytical methods for the same constraints as in CPIC, induce linear constraints on joint accelerations and torques through the robot dynamics. At runtime, an instantaneous QP enforces these viability-derived constraints on top of a passive DS-based torque controller, ensuring that the robot's state remains in the viable set over infinite time.

Inspired by the previous research on safety-critical control for robotic manipulators, and the potential of viability theory to provide forward-looking safety guarantees, in this project, we build on VPP-TC [6] and extend it from a fixed-base manipulator to a 9-DoF mobile base manipulator composed of a 2-DoF omnidirectional base and a 7-DoF arm. We employ the same viability-based formulation for self-collision, external collision, and joint limits, but now in a higher-dimensional state space, and compare against CPIC as a CBF-QP baseline with identical constraint objectives. Our results are consistent with the findings in [6]: the viability-based VPP-TC achieves substantially higher control-loop rates, improved numerical stability, and smoother behavior than the Hessian-dependent CBF-QP baseline, while guaranteeing safe, collision-free tracking for both the base and arm.

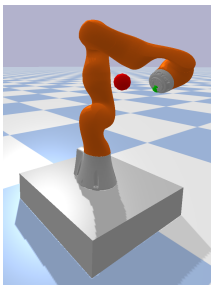


Fig. 1. Our 9 DOF mobile base manipulator research platform with 7 DOFs in arm and two prismatic joints on the base for tracking in planar x-y direction.

II. Preliminaries

A. Passive Torque Control with Dynamic Systems

We employ a Dynamic System (DS) to generate the nominal motion behavior for the robot. Let the desired task-space or configuration-space behavior be governed by a first-order autonomous differential equation:

$$\dot{\xi} = f(\xi), \quad \xi \in \mathbb{R}^m \tag{1}$$

where ξ represents the state (e.g., joint configuration in our project) and $f(\cdot)$ is a continuous vector field ensuring global asymptotic stability to a target.

To track this desired behavior passively, we utilize a passivity-based controller. As described in [1], the nominal

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control torque τ_{nom} is designed to dissipate energy with respect to the desired velocity field, ensuring that the closed-loop system behaves as a passive mechanical system. A typical formulation for the nominal torque is:

$$\tau_{nom} = -D(\dot{q} - J^{\dagger}\dot{\xi}_{des}) + G(q) \tag{2}$$

where D is a positive definite damping matrix, $J^{\dagger}$ is the pseudo-inverse of the Jacobian, and $G(q)$ compensates for gravity. This structure guarantees that the interaction remains safe and stable in the absence of active safety constraints.

B. Control Barrier Functions

For safety constraints defined on the full state $x = [q^{\top}, \dot{q}^{\top}]^{\top}$ (e.g., joint velocity limits or general state viability), let the safe set be defined by a scalar function $h_1(q, \dot{q}) \geq 0$. Since the control input (torque τ via acceleration $\ddot{q}$) appears explicitly in the first time derivative of h_1, we enforce the first-order condition:

$$\dot{h}_1(q, \dot{q}) + \alpha(h_1(q, \dot{q})) \geq 0 \tag{3}$$

Expanding $\dot{h}_1$ using the chain rule:

$$\frac{\partial h_1}{\partial q}\dot{q} + \frac{\partial h_1}{\partial \dot{q}}\ddot{q} + \alpha(h_1) \geq 0 \tag{4}$$

By substituting the robot dynamics $\ddot{q} = M^{-1}(\tau - C\dot{q} - G)$, this inequality becomes linear in the control torque τ.

For safety constraints defined solely on the configuration space (e.g., collision avoidance), let the safe set be $h_2(q) \geq 0$. The control input does not appear in $\dot{h}_2$, requiring a second-order (or high-order) CBF condition:

$$\ddot{h}_2(q, \dot{q}) + \gamma_1 \dot{h}_2(q, \dot{q}) + \gamma_2 h_2(q) \geq 0 \tag{5}$$

where $\gamma_1, \gamma_2 > 0$ are design gains. The second derivative is given by:

$$\ddot{h}_2 = \frac{\partial h_2}{\partial q}\ddot{q} + \dot{q}^{\top}\frac{\partial^2 h_2}{\partial q^2}\dot{q} \tag{6}$$

Similar to the first-order case, substituting $\ddot{q}$ yields a linear constraint on τ.

Both formulations can be unified into a standard QP constraint form:

$$A_{cbf}(x)\tau \leq b_{cbf}(x) \tag{7}$$

This linear constraint serves as the safety filter in the baseline CPIC method, strictly enforcing safety by modifying the nominal torque.

C. Viability

In the n-DoF mobile base manipulator case, the state is defined as $(q, \dot{q})$ in the feasible set $\mathcal{F} \in \mathbb{R}^{2n}$, where $q \in \mathbb{R}^n$ is the joint positions and $\dot{q} \in \mathbb{R}^n$ is the joint velocities. A state $(q, \dot{q})$ is considered *viable*, from that state, there exists a control input sequence that guarantees future states satisfying all constraints [7]. Accordingly, we define the viability set $\mathcal{V}$ gathering all viable states as:

$$\begin{aligned}(q(0), \dot{q}(0)) \in \mathcal{V} \iff \exists (\ddot{q}_i)_{i=0}^{\infty} : (q(t), \dot{q}(t)) \in \mathcal{F};\ \forall t \geq 0,\\ |\ddot{q}_i| \leq \ddot{q}^{\max};\ \forall i \geq 0.\end{aligned} \tag{8}$$

According to the definition of the viability set $\mathcal{V}$, we define the safe set in our task as $\mathcal{V}$.

III. Problem Formulation and Methods

The dynamic of the 7-DoF manipulator with the 3-DoF mobile base is derived by Euler-Lagrange equation as follows:

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau_c + \tau_{\text{ext}}, \tag{9}$$

A. Viability-Preserving Passive Torque Control (VPP-TC)

The goal of our project is to provide torque bounds for nominal passive torque controllers on a mobile-base manipulator. To ensure safe control, we consider three categories of constraints when computing the overall viability set. These include the self-collision-free viability, the external-collision-free viability, and the joint-limit viability, formally defined as follows:

1) Self-collision-free viability:

$$\mathcal{V}_{\text{sca}} := \{(q_0, \dot{q}_0) \mid q(t) \in \mathcal{C}_{\text{sca}},\ \forall t \in [0, T_{\text{br}}]\}. \tag{10}$$

2) External-collision-free viability:

$$\mathcal{V}_{\text{eca}} := \{(q_0, \dot{q}_0) \mid q(t) \in \mathcal{C}_{\text{eca}},\ \forall t \in [0, T_{\text{br}}]\}. \tag{11}$$

3) Joint-limit viability:

$$V_{\text{jnt}} := \{(q, \dot{q}) \mid \ddot{q}_{\text{lb}} < \ddot{q} < \ddot{q}_{\text{ub}}\} \tag{12}$$

where $\mathcal{C}_{\text{sca}}$ and $\mathcal{C}_{\text{eca}}$ are sets of self-collision-free and external-collision-free configurations. In practice, we used the empirical surrogate of the viability set for the self-/external-collision free viability instead, to learn the set from data.

1) Self-collision-free viability:

$$\hat{V}_{\text{sca}} := \{(q, \dot{q}) \mid \Gamma(q, \dot{q}) > 0\} \tag{13}$$

2) External-collision-free viability:

$$\hat{V}_{\text{eca}} := \{(q, \dot{q}) \mid S_v(p, q, \dot{q}) > 0\} \tag{14}$$

Concretely, we learn $\hat{V}_{\text{sca}}$ through a transformer-based neural network, which maps the input state $(q, \dot{q})$ to $\gamma(q, \dot{q}) : \mathbb{R}^{20} \to \mathbb{R}^1$. When the $\Gamma(q, \dot{q}) \in (0, \epsilon_{\text{sca}})$, self-collision avoidance constraint will be activated by enforcing $\Delta\Gamma(q, \dot{q}) \geq 0$. Additionally, we set $\gamma_{\text{threshold}} = \epsilon_{\text{sca}}$ via a threshold sweep over Γ, yielding the best recall for the viable class. Using a Taylor expansion and neglecting higher-order terms, we obtain

$$\begin{aligned}\Delta\Gamma(q, \dot{q}) &= \nabla_q \Gamma^{\top}\, \Delta q + \nabla_{\dot{q}} \Gamma^{\top}\, \Delta\dot{q} \\ &= \nabla_q \Gamma^{\top} \left(\dot{q}\, \delta t + \tfrac{1}{2}\ddot{q}\, \delta t^2\right) + \nabla_{\dot{q}} \Gamma^{\top} \left(\ddot{q}\, \delta t\right) \\ &= \left(\tfrac{1}{2}\nabla_q \Gamma^{\top}\, \delta t^2 + \nabla_{\dot{q}} \Gamma^{\top}\, \delta t\right) \ddot{q} + \nabla_q \Gamma^{\top}\, \dot{q}\, \delta t,\end{aligned} \tag{15}$$

Then define $g_{se} = \frac{1}{2}\nabla_q \Gamma^{\top}\, \delta t^2 + \nabla_{\dot{q}} \Gamma^{\top}\, \delta t.$ and $b_s = \nabla_q \Gamma^{\top}\, (\dot{q}\, \delta t)$, together with the system dynamics Eq.(9), we can derive the self-collision avoidance constraint as

$$g_{se} M^{-1} \tau \ \geq\ g_{se} M^{-1}(C\dot{q} + G) - b_s \tag{16}$$

For $\hat{V}_{\text{eca}}$, we make use of Bernstein polynomials to encode the viability set. We use this polynomial representation [8] to learn a signed distance function (SDF) for each link in the

robot. With SDFs learnt for each link separately, we define a global SDF as the pointwise minimum of all the link SDFs:

$$S(p,q) = \min_{n=1,\dots,N} S_n(p,q) \tag{17}$$

The metric that we are interested in keeping positive is $S_v(p,q) = \min_{t\in[0,T_{br}]} S(p,q)$, which is the minimum distance value during the braking trajectory to any link. We enforce this constraint through enforcing $\Delta S_v(p,q,\dot{q}) > 0$ using taylor expansion:

$$\begin{aligned}\Delta S_v(p,q,\dot{q}) &= \nabla_q S_v^\top \,\Delta q + \nabla_{\dot{q}} S_v^\top \,\Delta\dot{q} + \nabla_p S_v^\top \,\Delta p \\ &= \nabla_q S_v^\top \left(\dot{q}\,\delta t + \tfrac{1}{2}\ddot{q}\,\delta t^2\right) + \nabla_{\dot{q}} S_v^\top \,(\ddot{q}\,\delta t) \\ &\quad + \nabla_p S_v^\top \,\Delta p \\ &= \left(\tfrac{1}{2}\nabla_q S_v^\top \,\delta t^2 + \nabla_{\dot{q}} S_v^\top \,\delta t\right)\ddot{q} + \nabla_q S_v^\top \,\dot{q}\,\delta t, \\ &\quad + \nabla_p S_v^\top \,\Delta p\end{aligned} \tag{18}$$

Then define $g_{ee} = \frac{1}{2}\nabla_q S_v^\top \,\delta t^2 + \nabla_{\dot{q}} S_v^\top \,\delta t.$ and $b_s = \nabla_q S_v^\top \,(\dot{q}\,\delta t) + \nabla_p S_v^\top \,\Delta p$. We can derive the external-collision avoidance constraint as

$$g_{ee} M^{-1}\tau \;\geq\; g_{ee} M^{-1}(C\dot{q} + G) - b_e \tag{19}$$

As a result, we can formulate the optimization problem with the above constraints below.

$$\begin{aligned}&\min_{\tau}\; 0.5\left\|\hat{J}(q)^{-\top}\tau - F_c(x)\right\|^2 \;+\; \alpha\|\tau\|^2 \\ &\text{s.t.}\quad \text{Constraints (16,19,12) are all satisfied.} \\ &\qquad \tau \in [\tau_{\min}, \tau_{\max}]\end{aligned} \tag{20}$$

B. Baseline: CBFs method in CPIC [1]

In CPIC, the authors utilized the following Control Barrier Functions with respect to three constraints.

1) Self-Collision Avoidance Constraint

$$\begin{aligned}\nabla_q h_{\text{SCA}}(q)\,\ddot{q} \;&\geq\; b_{\text{SCA}}(q) \\ h_{\text{SCA}}(q) &= \Gamma_{\text{SCA}}(q) - \varepsilon_{\text{SCA}}, \\ b_{\text{SCA}}(q) &= -k_1\, h_{\text{SCA}}(q) - k_2\,\nabla_q h_{\text{SCA}}(q)\,\dot{q} \\ &\quad - \dot{q}^\top \nabla_q^2 h_{\text{SCA}}(q)\,\dot{q}\end{aligned} \tag{21}$$

2) External Collision Avoidance Constraint

$$\begin{aligned}\nabla_q h^{(i)}_{\text{ECA}}(q)\,\ddot{q} \;&\geq\; b^{(i)}_{\text{ECA}}(q), \\ h^{(i)}_{\text{ECA}}(q) &= \Gamma^{(i)}_{\text{SDF}}(q, x_0) - \varepsilon_{\text{SDF}}, \\ b^{(i)}_{\text{ECA}}(q) &= -k_1\, h^{(i)}_{\text{ECA}}(q) - k_2\,\nabla_q h^{(i)}_{\text{ECA}}(q)\,\dot{q} \\ &\quad - \dot{q}^\top \nabla_q^2 h^{(i)}_{\text{ECA}}(q)\,\dot{q}.\end{aligned} \tag{22}$$

3) Joint Limit Avoidance Constraint
It remains the same as ours Eq.(12).

IV. Experiments and Results

To validate the performance of the VPP-TC control framework on our robot model, we conducted four setpoint tracking experiments using the PyBullet simulation environment. These variations were designed to investigate the controller's behavior across typical operational scenarios. The objective function is taken from Eq.(20), which implements the task.

We compare the performance of two distinct methods: Viability Set Constraints [6] and Control Barrier Function (CBF) constraints [1]. Key metrics of comparison are jerkiness of the trajectory and the control loop rate for each method.

The end-effector jerk is computed as:

$$j(t) = \left\|\frac{d^3\mathbf{x}}{dt^3}\right\| = \|\dddot{\mathbf{x}}(t)\| \tag{23}$$

where $\mathbf{x}(t) \in \mathbb{R}^3$ denotes the end-effector position in Cartesian space. The jerk magnitude is quantified using the root-mean-square (RMS) metric:

$$j_{\text{RMS}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} \|\dddot{\mathbf{x}}(t_i)\|^2} \tag{24}$$

Our investigation proceeds in two phases:

- Fixed Base Analyses: We fix the base and compare the performance of VPP-TC and CBF constraints
- Moving Base Analysis: We actuate the base (only linear velocity in orthogonal directions, without any yaw rate) and implement VPP-TC constraints on the 9 DOF system (7 from arm and 2 from movable base)

A. Fixed Base Experiments

1) Obstacle before Target Point: Setup: An obstacle is kept between the arm and the target point, close to the target point. Video of the experiment with VPP-TC constraints: *Link to VPP-TC Constraint Experiment*. Video of experiment with CBF constraints: *Link to CBF Constraint Experiment*

Observations: VPP-TC constraints avoid the obstacle and converge to the target point in about 3 seconds. It maintains a minimum threshold distance γ_{eca} set at 0.08 m. The jerk metric j_{VPP} is 1413 m/s³.

The CBF constraints takes 9 seconds to converge to the target point. Its motion is jerky and oscillates around the target point. The jerk metric j_{CBF} is 14974 m/s³, substantially more than the viability constraints. It is more conservative than VPP-TC in maintaining distance from the obstacle for same distance threshold set.

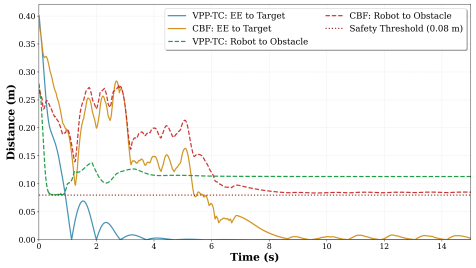


Fig. 2. Comparison of VPP-TC constraints and CBF constraints for the case with obstacle in between. The VPP-TC converges faster and is smoother. The CBF constraints result in a jerky motion

2) Target coincident with Obstacle: Setup: The target point is kept coincident with the external obstacle. Video of the experiment: *Link to Task with VPP-TC Constraints*

Observations: VPP-TC constraints approach the target but try to avoid the obstacle at the same place. It eventually converges to a local minima where it maintains a constant distance from the target point such that it stays outside the distance threshold from obstacles. The jerk observed j_{VPP} was 2584 m/s³.

The CBF constraints are not able to converge to a stable position but do still maintain a minimum distance threshold from the obstacle while approaching the target. The observed jerk j_{CBF} was 15189 m/s³.

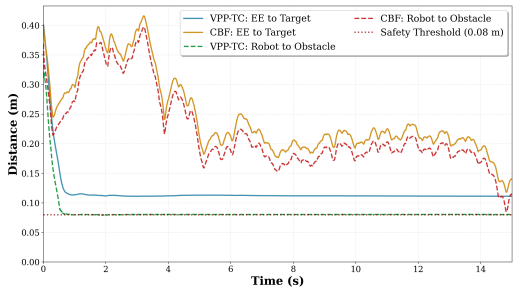


Fig. 3. Comparison of VPP-TC constraints and CBF constraints for coincident obstacle and target. VPP-TC converges to a constant distance from the target. The CBF constraints get closer but keep jerking around.

3) Target inside Robot Link (Self-Collision Check): Setup: To test the self-collision avoidance, we keep the target point inside the base at its edges. Then we sampled 8 million of the data points as our dataset for $\Gamma(q, \dot{q})$ training. We trained for 60 epochs, with 99.69% accuracy and 99.72% recall. Video of the experiment: *Link to Self Collision Avoidance with VPP-TC constraints*

Observations: Both VPP-TC and CBF constraints take the end effector closer to the target point and stabilize at a threshold distance from the base to avoid self-collision. Calculated jerks are j_{VPP} 1724 m/s³ and j_{CBF} 13158 m/s³. However, we still observe the parameter-sensitive behavior of the VPP-TC, which requires cautious parameter choice of $\gamma_{\text{threshold}}$, which won't be decided solely by sweeping over Γ. Also, the performance of the Transformer-based model showed some instability, for example, exhibiting sensitivity to the size of the training dataset and the number of training epochs.

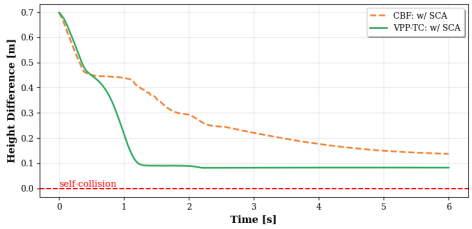


Fig. 4. Comparison of VPP-TC constraints and CBF constraints for the self-collision case. VPP-TC converges to a constant distance from the target. The CBF constraints get closer but take more time to converge.

B. Mobile Base Experiments

1) All Three Constraints Activated: Setup: We extend all the viability constraints to the case of additional 2 DOFs for an actuated base. The base only has linear velocities in orthogonal directions and no angular velocity. Video of the experiment: *Link to Moving base with all viability constraints*

Observations: We extended only the VPP-TC constraints to actuated base. The base and arm moves to reach the target point but due to local decision making, the arm frequently assumed a singular configuration.

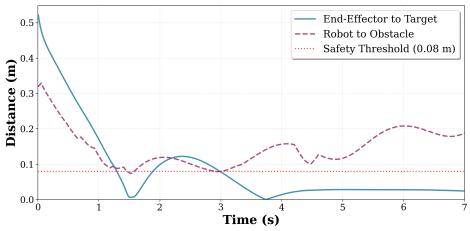


Fig. 5. All three VPP-TC constraints implemented on mobile base arm.

Control Loop Rate: For an experiment with all constraints active for both viability constraints and CBF constraints, we calculated the control loop rate as 110 Hz for VPP-TC, but 47 Hz for CBF constraints, considerably slower. We observed that the bottleneck in the CBF was the hessian calculation.

Jerkiness: We observe consistently, both visually and numerically, that the VPP-TC controller results in a smoother trajectory, and CBF constraints display some jerk in the motion. We hypothesize that this happens due to the numerical instability in hessian calculation in the algorithm.

V. Conclusion

We observed that the viability set formulation of these safety constraints is a viable approach with considerable advantages over Control Barrier Functions (CBF) based methods.

The nature of formulation of viability sets in our work extends it the following properties:

- **Predictive**: Because the method pre-calculates viability sets by simulating trajectories into the future, it has predictive guarantee of safety as compared to only reactive notion of safety in CBF constraints. This leads to a quicker convergence.
- **Faster**: VPP-TC calculates gradients only up to first order as compared to CBF constraints which calculate hessians. This results in a faster solve time and therefore a faster control loop rate
- **Jerkiness**: Because of the lack of higher order gradient calculations in VPP-TC, it demonstrates smoother trajectories compared to CBF constraint based QP controller.

We plan to extend the work in future by:

- Applying global planning to VPP-TC formulation for mobile base manipulator to avoid singularities and formulating constraints that avoid singular configurations in the optimization problem.
- Improving the stability of current VPP-TC performance.
- Applying these constraints as rewards in an RL formulation [9].

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